

MMICT&BM	MM INSTITUTE OF COMPUTER TECHNOLOGY & BUSINESS MANAGEMENT-MULLANA	LABORATORY MANUAL
	PRACTICAL EXPERIMENT INSTRUCTION SHEET	
	Title: Install R/R Studio, explore R Studio interface, run basic commands. Write a program to display "Welcome to R Programming".	
EXPERIMENT NO.: BCA- 602	ISSUE No. : 01	ISSUE DATE:
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Aim: Installation and Setup of R/R Studio.

- Install R/R Studio, explore R Studio interface, run basic commands.

Write a program to display "Welcome to R Programming".

Solution:

Installing R and RStudio on Windows

To Install R and R Studio on Windows we will have to download R and R Studio with the following steps.

- First, you need to set up an R environment in your local machine. Visit the official CRAN website:

1: Install R

RStudio requires R 3.6.0+. Choose a version of R that matches your computer's operating system.

R is not a Posit product. By clicking on the link below to download and install R, you are leaving the Posit website. Posit disclaims any obligations and all liability with respect to R and the R website.

DOWNLOAD AND INSTALL R

2: Install RStudio

DOWNLOAD RSTUDIO DESKTOP FOR WINDOWS

Size: 281.24 MB | SHA-256: 3A553330 | Version: 2025.05.1+51 | Released: 2025-06-05

- You have to download both the applications first go with R Base and then install RStudio. after click on install R you will get a new page like this.

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The Comprehensive R Archive Network

Download and Install R

Precompiled binary distributions of the base system and contributed packages, **Windows and Mac** users most likely want one of these versions of R.

- Download R for Linux (Debian, Fedora, Redhat, Ubuntu)
- Download R for macOS
- Download R for Windows

R is part of many Linux distributions, you should check with your Linux package management system in addition to the link above.

Source Code for all Platforms

Windows and Mac users most likely want to download the precompiled binaries listed in the upper box, not the source code. The sources have to be compiled before you can use them. If you do not know what this means, you probably do not want to do it!

- The latest release (2025-06-13, Great Square Root) [R-4.5.1 tar.gz](#), read [what's new](#) in the latest version.
- The CRAN directory [src/base-pre-release](#) contains R alpha, beta, and rc releases as daily snapshots in time periods before a planned release.
- Between releases, the same directory [src/base-pre-release](#) contains snapshots of current patched and development versions. Please read about [new features](#) and [bug fixes](#) before filing corresponding feature requests or bug reports.
- Alternatively, daily snapshots are [available here](#).
- Source code of older versions of R is [available here](#).
- Contributed extension [packages](#)

Questions About R

- If you have questions about R like how to download and install the software, or what the license terms are, please read our [answers to frequently asked questions](#) before you send an email.

Supporting CRAN

- CRAN operations, most importantly hosting, checking, distributing, and archiving of R add-on packages for various platforms, crucially rely on technical, emotional, and financial support by the R community.

Please consider making [financial contributions](#) to the R Foundation for Statistical Computing.

What are R and CRAN?

R is 'GNU S', a freely available language and environment for statistical computing and graphics which provides a wide variety of statistical and graphical techniques: linear and nonlinear modelling, statistical tests, time series analysis, classification, clustering, etc. Please consult the [R project homepage](#) for further information.

CRAN is a network of ftp and web servers around the world that store identical, up-to-date, versions of code and documentation for R. Please use the CRAN [mirror](#) nearest to you to minimize network load.

Submitting to CRAN

Download R

- Here we can select the linux, mac or windows any one according to users system. you have to click on for which you want to install.

R-4.5.1 for Windows

Download R-4.5.1 for Windows (56 megabytes, 64 bit)

[Release on the Windows binary distribution](#)
[New features in this version](#)

This build requires UCRT, which is part of Windows since Windows 10 and Windows Server 2016. On older systems, UCRT has to be installed manually from [here](#).

If you want to double-check that the package you have downloaded matches the package distributed by CRAN, you can compare the [md5sum](#) of the .exe to the [fingerprint](#) on the master server.

Frequently asked questions

- Does R run under my version of Windows?
- How do I update packages in my previous version of R?

Please see the [R FAQ](#) for general information about R and the [R Windows FAQ](#) for Windows-specific information.

Other builds

- Patches to this release are incorporated in the [patched snapshot build](#).
- A build of the development version (which will eventually become the next major release of R) is available in the [development snapshot build](#).
- [Previous releases](#)

Note to webmasters: A stable link which will redirect to the current Windows binary release is
`<CRAN_MIRROR>-bin/windows/base/release.html`

Last change: 2025-06-13

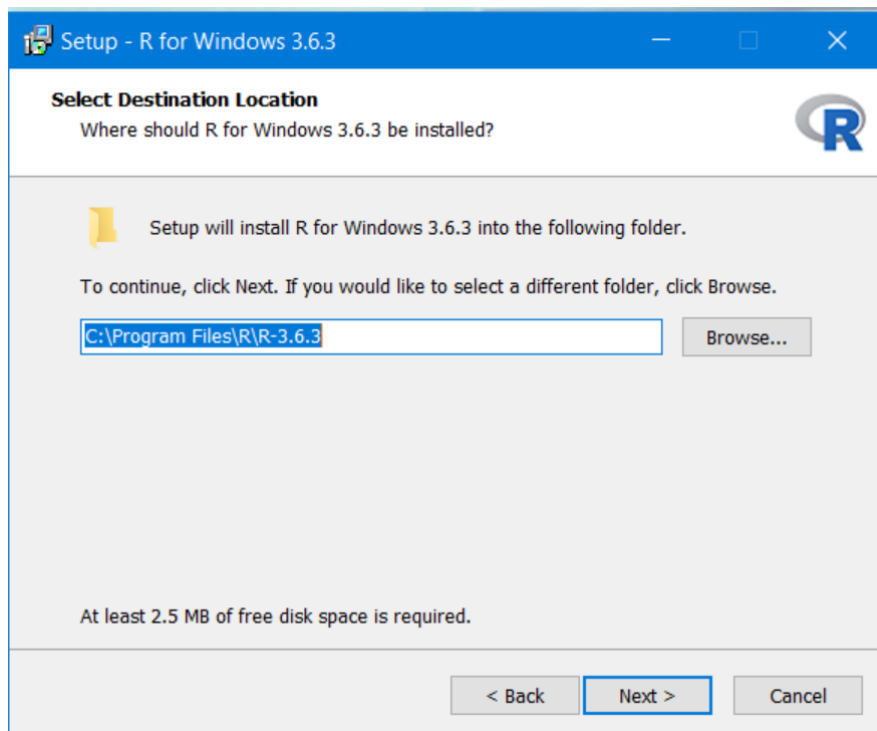
Install R

- now click on the link show above in image so R base start downloading and after again go to main page and download and click on Install RStudio.

Install R and R Studio Using the Installation Wizard

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Step 1: After downloading R for the Windows platform, install it by double-clicking it.

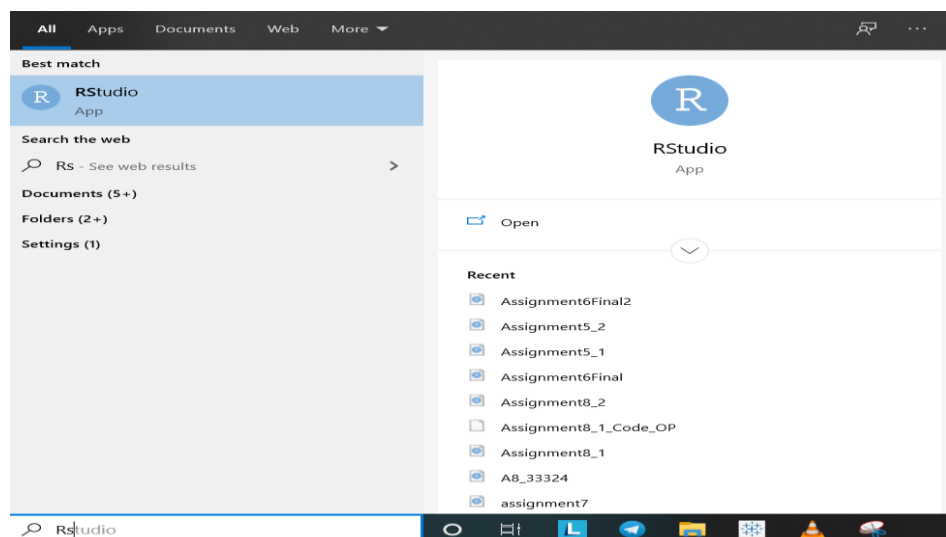


Step 2: Download R Studio from their official page. Note: It is free of cost (under AGPL licensing).

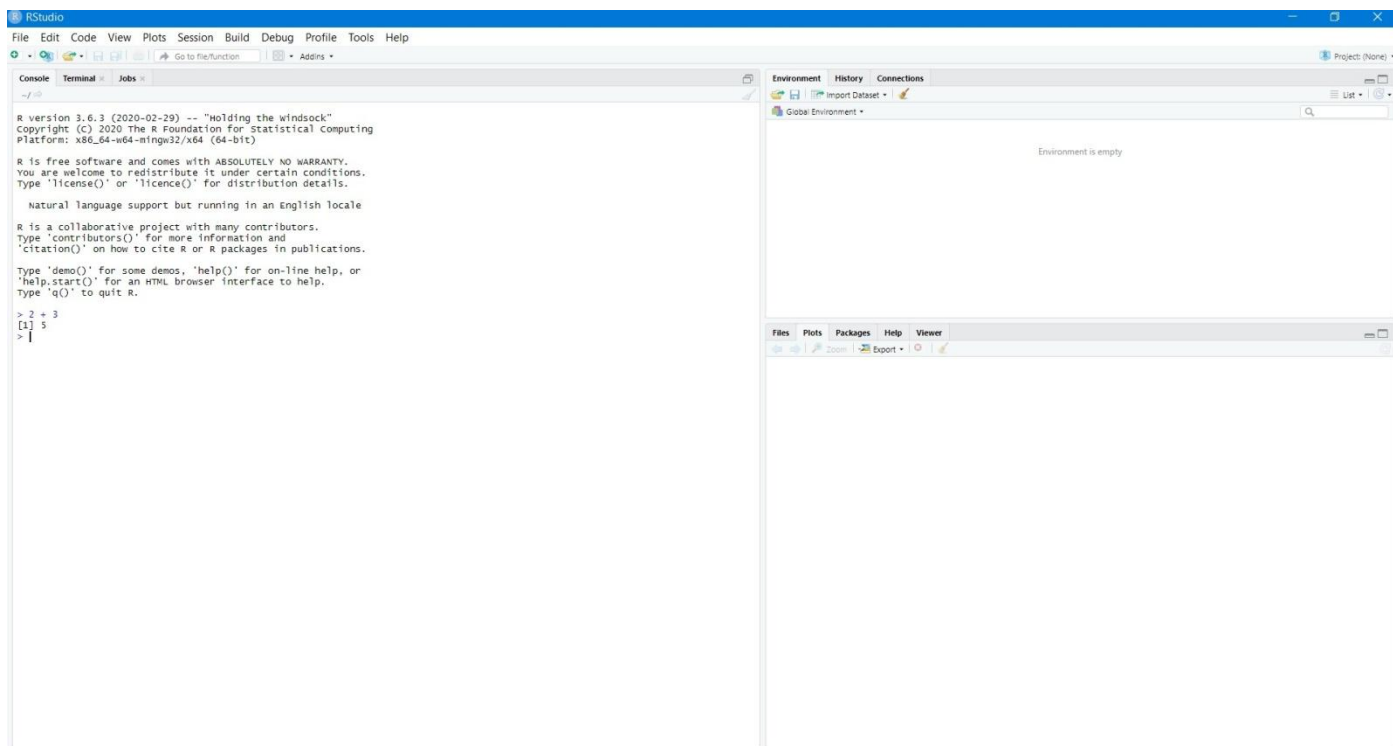
Step 3: After downloading, you will get a file named "RStudio-1.x.xxxx.exe" in your Downloads folder.

Step 4: Double-click the installer, and install the software.

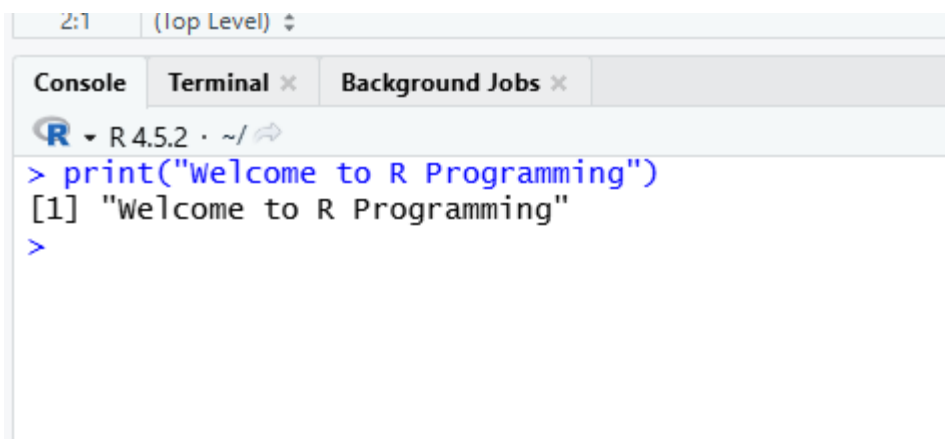
Step 5: Search for RStudio in the Window search bar on Taskbar



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Step 6: Your installation is successful.**Solution:**

```
print("Welcome to R Programming")
```

Output:

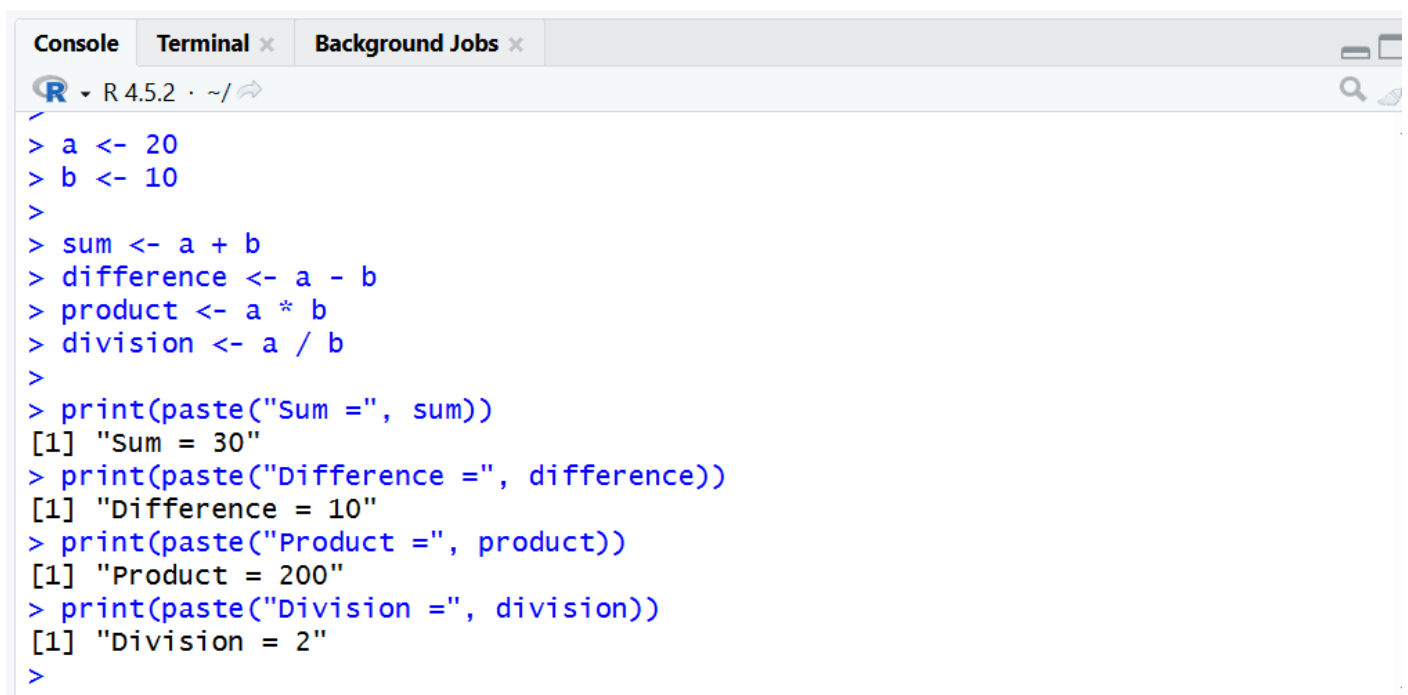
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Aim: Write an R script to calculate the sum, difference, product, and division of two numbers.

Solution:

```
a <- 20
b <- 10
sum <- a + b
difference <- a - b
product <- a * b
division <- a / b
print(paste("Sum =", sum))
print(paste("Difference =", difference))
print(paste("Product =", product))
print(paste("Division =", division))
```

Output:



```
Console Terminal x Background Jobs x
R 4.5.2 · ~/
> a <- 20
> b <- 10
>
> sum <- a + b
> difference <- a - b
> product <- a * b
> division <- a / b
>
> print(paste("Sum =", sum))
[1] "Sum = 30"
> print(paste("Difference =", difference))
[1] "Difference = 10"
> print(paste("Product =", product))
[1] "Product = 200"
> print(paste("Division =", division))
[1] "Division = 2"
>
```

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Aim: Variables and Operators in R

- Perform arithmetic, relational, and logical operations using variables.

Experiment: Create variables of different data types in R and display their type using appropriate functions

Solution:

```
# Arithmetic Operations
```

```
a <- 10
```

```
b <- 5
```

```
add <- a + b
```

```
sub <- a - b
```

```
mul <- a * b
```

```
div <- a / b
```

```
print(add)
```

```
print(sub)
```

```
print(mul)
```

```
print(div)
```

```
# Relational Operations
```

```
print(a > b)
```

```
print(a < b)
```

```
print(a == b)
```

```
print(a != b)
```

```
# Logical Operations
```

```
x <- TRUE
```

```
y <- FALSE
```

```
print(x & y)
```

```
print(x | y)
```

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```
print(!x)
```

```
# Variables of Different Data Types
```

```
num <- 25
```

```
decimal <- 10.5
```

```
text <- "R Programming"
```

```
flag <- TRUE
```

```
print(typeof(num))
```

```
print(typeof(decimal))
```

```
print(typeof(text))
```

```
print(typeof(flag))
```

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Output:

```

SOURCE
Console Terminal Background Jobs
R - R4.5.2 - ~/
> # Arithmetic Operations
> a <- 10
> b <- 5
>
> add <- a + b
> sub <- a - b
> mul <- a * b
> div <- a / b
>
> print(add)
[1] 15
> print(sub)
[1] 5
> print(mul)
[1] 50
> print(div)
[1] 2
>
> # Relational Operations
> print(a > b)
[1] TRUE
> print(a < b)
[1] FALSE
> print(a == b)
[1] FALSE
> print(a != b)
[1] TRUE
>
> # Logical Operations
> x <- TRUE
> y <- FALSE
>
> print(x & y)
[1] FALSE
> print(x | y)
[1] TRUE
> print(!x)
[1] FALSE
>
> # Variables of Different Data Types
> num <- 25
> decimal <- 10.5
> text <- "R Programming"
> flag <- TRUE
>
> print(typeof(num))
[1] "double"
> print(typeof(decimal))
[1] "double"
> print(typeof(text))
[1] "character"
> print(typeof(flag))
[1] "logical"
> |

```


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Aim: Vectors and Vectorized Operations- Create vectors and apply vectorized arithmetic operations.

Experiment:

1. Write a R program to perform following operations on Vector: min(), max(), mean(), sqrt(), length(), sum(), prod(), sort()(in ascending and descending order), rev(), addition/subtraction/multiplication/division of two vectors.
2. Write a R program to create a list containing a vector, a matrix and a list and perform following operations:
 - i. Update the elements in the list.
 - ii. Merge two lists into one list.
 - iii. Count number of objects in a given list.

Solution:

```
v1 <- c(2, 4, 6, 8, 10)
```

```
v2 <- c(1, 3, 5, 7, 9)
```

```
min(v1)
```

```
max(v1)
```

```
mean(v1)
```

```
sqrt(v1)
```

```
length(v1)
```

```
sum(v1)
```

```
prod(v1)
```

```
sort(v1)
```

```
sort(v1, decreasing=TRUE)
```

```
rev(v1)
```

```
v1 + v2
```

```
v1 - v2
```

```
v1 * v2
```

Form No. MMEC-FRM-32 B (Rev. No. 00)

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```
vec <- c(10, 20, 30)
```

```
mat <- matrix(1:6, nrow=2)
```

```
sublist <- list("A", "B", "C")
```

```
my_list <- list(vec, mat, sublist)
```

```
my_list[[1]][2] <- 99
```

```
my_list[[2]][1,1] <- 50
```

```
my_list[[3]][2] <- "Z"
```

```
list2 <- list(100, 200, 300)
```

```
merged_list <- c(my_list, list2)
```

```
length(my_list)
```

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Output:

```

Console Terminal Background Jobs
R - R 4.5.2 - ~/
> v1 <- c(2, 4, 6, 8, 10)
> v2 <- c(1, 3, 5, 7, 9)
> min(v1)
[1] 2
> max(v1)
[1] 10
> mean(v1)
[1] 6
> sqrt(v1)
[1] 1.414214 2.000000 2.449490 2.828427 3.162278
> length(v1)
[1] 5
> sum(v1)
[1] 30
> prod(v1)
[1] 3840
> sort(v1)
[1] 2 4 6 8 10
> sort(v1, decreasing=TRUE)
[1] 10 8 6 4 2
> rev(v1)
[1] 10 8 6 4 2
> v1 + v2
[1] 3 7 11 15 19
> v1 - v2
[1] 1 1 1 1 1
> v1 * v2
[1] 2 12 30 56 90
> v1 / v2
[1] 2.000000 1.333333 1.200000 1.142857 1.111111
> vec <- c(10, 20, 30)
> mat <- matrix(1:6, nrow=2)
> sublist <- list("A", "B", "C")
> my_list <- list(vec, mat, sublist)
> my_list[[1]][2] <- 99
> my_list[[2]][1,1] <- 50
> my_list[[3]][2] <- "Z"
> list2 <- list(100, 200, 300)
> merged_list <- c(my_list, list2)
> length(my_list)
[1] 3
>

```

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Aim: Subsetting Data in R

Subset vectors using indexing, logical conditions, and named elements.

Experiment: Create a vector of 10 elements and extract even-indexed elements.

Solution:

```
v <- c(5, 10, 15, 20, 25, 30, 35, 40, 45, 50)

v[1]

v[3]

v[c(1, 3, 5)]

v[v > 25]

names(v) <- c("a","b","c","d","e","f","g","h","i","j")

v["d"]

v[seq(2, length(v), by = 2)]
```

Output:

```
9:1 (Top Level)
Console Terminal Background Jobs
R 4.5.2 ~/
> v <- c(5, 10, 15, 20, 25, 30, 35, 40, 45, 50)
> v[1]
[1] 5
> v[3]
[1] 15
> v[c(1, 3, 5)]
[1] 5 15 25
> v[v > 25]
[1] 30 35 40 45 50
> names(v) <- c("a","b","c","d","e","f","g","h","i","j")
> v["d"]
d
20
> v[seq(2, length(v), by = 2)]
b d f h j
10 20 30 40 50
> |
```

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Aim: Handling NA and NULL Values

Detect, remove, and replace NA and NULL values; apply coding standards.

Experiment: Create a vector with NA values and write a program to remove them.

Solution:

```
v <- c(10, 20, NA, 30, NA, 40)
```

```
is.na(v)
```

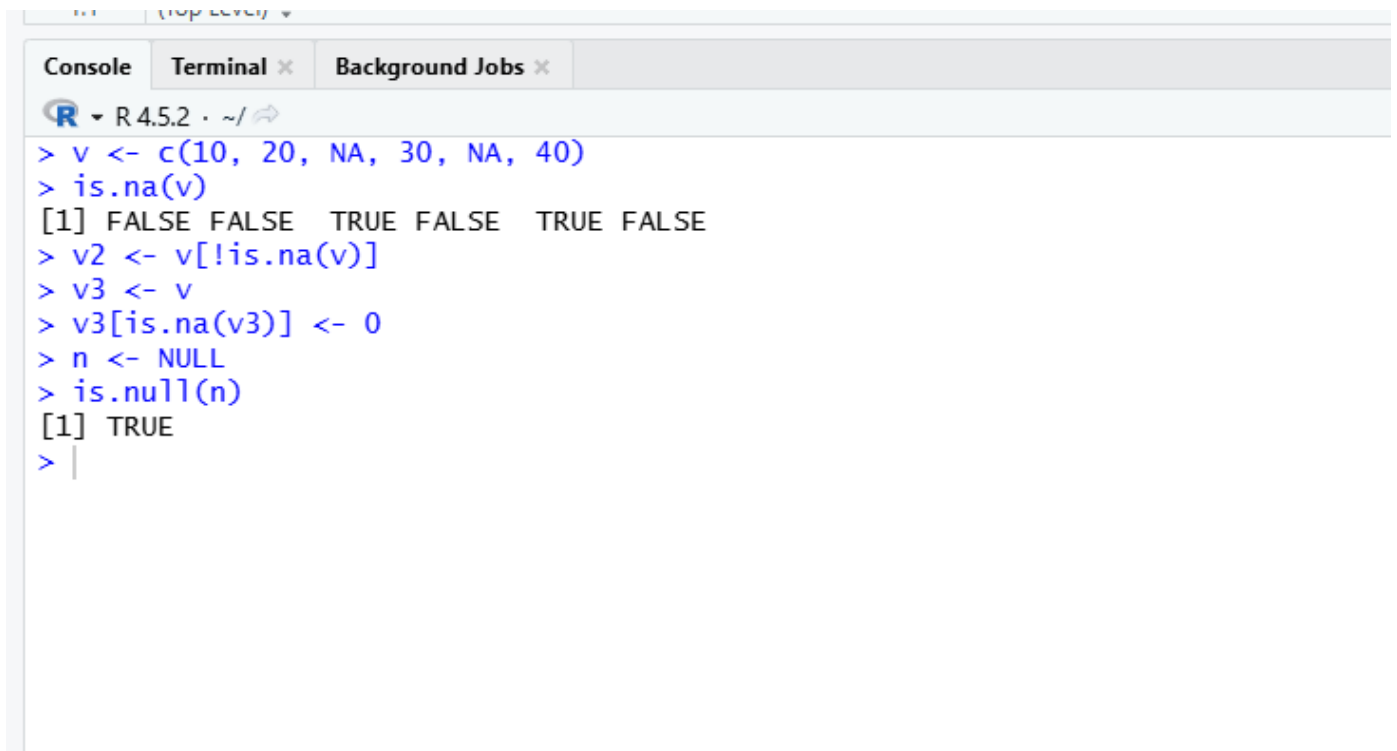
```
v2 <- v[!is.na(v)]
```

```
v3 <- v
```

```
v3[is.na(v3)] <- 0
```

```
n <- NULL
```

```
is.null(n)
```

Output:


```

> v <- c(10, 20, NA, 30, NA, 40)
> is.na(v)
[1] FALSE FALSE  TRUE FALSE  TRUE FALSE
> v2 <- v[!is.na(v)]
> v3 <- v
> v3[is.na(v3)] <- 0
> n <- NULL
> is.null(n)
[1] TRUE
>

```

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Aim: Conditional Statements Using if-else and Nested if-else Statements

Implement decision-making using if statements.

Experiment: Write a program to check leap year in R.

Solution:

```

year <- 2024

if (year %% 400 == 0) {

  cat("Leap Year")

} else if (year %% 100 == 0) {

  cat("Not a Leap Year")

} else if (year %% 4 == 0) {

  cat("Leap Year")

} else {

  cat("Not a Leap Year")

}

```

Output:

2/11 (top level) ▾

Console

Terminal ×

Background Jobs ×

 R 4.5.2 · ~/ 

```
+ year <- 2024
+ if (year %% 400 == 0) {
+   cat("Leap Year")
+ } else if (year %% 100 == 0) {
+   cat("Not a Leap Year")
+ } else if (year %% 4 == 0) {
+   cat("Leap Year")
+ } else {
+   cat("Not a Leap Year")
+ }
+
+ Leap Year|
```

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Aim: For Loop Implementation and While Loop Implementation**Use for loop for repetitive tasks and sequence generation.****Experiment:** 1. Write a user defined function to generate Fibonacci series using while loop.

2. Generate the multiplication table of a given number using a for loop.

3. Write a program in R to make a simple calculator.

Solution 1:

```
fibonacci <- function(n) {
```

```
  a <- 0
```

```
  b <- 1
```

```
  i <- 1
```

```
  while (i <= n) {
```

```
    cat(a, " ")
```

```
    c <- a + b
```

```
    a <- b
```

```
    b <- c
```

```
    i <- i + 1
```

```
  }
```

```
}
```

```
fibonacci(10)
```


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Output:

```

15:1 (Top Level)
Console Terminal Background Jobs
R 4.5.2 ~/

> fibonacci <- function(n) {
+   a <- 0
+   b <- 1
+   i <- 1
+   while (i <= n) {
+     cat(a, " ")
+     c <- a + b
+     a <- b
+     b <- c
+     i <- i + 1
+   }
+ }
>
> fibonacci(10)
0 1 1 2 3 5 8 13 21 34
>

```

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Solution:2

```

num <- as.integer(readline("Enter a number: "))

cat("Multiplication Table of", num, "\n")

for (i in 1:10) {

  cat(num, "x", i, "=", num * i, "\n")

}

```

Output:

The screenshot shows an R console window with the following content:

```

8:1 (Top Level)
Console Terminal Background Jobs
R 4.5.2 ~ /
> num <- as.integer(readline("Enter a number: "))
Enter a number: 5
> cat("Multiplication Table of", num, "\n")
Multiplication Table of 5
>
> for (i in 1:10) {
+   cat(num, "x", i, "=", num * i, "\n")
+ }
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50
> |

```

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Solution3:

```

num1 <- as.numeric(readline(prompt = "Enter the first number: "))
num2 <- as.numeric(readline(prompt = "Enter the second number: "))
op <- readline(prompt = "Enter an operation (+, -, *, /): ")
if (op == "+") {
  result <- num1 + num2
} else if (op == "-") {
  result <- num1 - num2
} else if (op == "*") {
  result <- num1 * num2
} else if (op == "/") {
  if (num2 == 0) {
    result <- "Error: Division by zero is not allowed"
  } else {
    result <- num1 / num2
  }
} else { result <- "Error: Invalid operation"}
cat("Result:", result, "\n")

```

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Output:

```

Console Terminal Background Jobs
R - R 4.5.2 - ~/
> num1 <- as.numeric(readline(prompt = "Enter the first number: "))
Enter the first number: 45
> num2 <- as.numeric(readline(prompt = "Enter the second number: "))
Enter the second number: 25
> op <- readline(prompt = "Enter an operation (+, -, *, /): ")
Enter an operation (+, -, *, /): +
>
> if (op == "+") {
+   result <- num1 + num2
+ } else if (op == "-") {
+   result <- num1 - num2
+ } else if (op == "*") {
+   result <- num1 * num2
+ } else if (op == "/") {
+   if (num2 == 0) {
+     result <- "Error: Division by zero is not allowed"
+   } else {
+     result <- num1 / num2
+   }
+ } else {
+   result <- "Error: Invalid operation"
+ }
>
> # Print the result
> cat("Result:", result, "\n")
Result: 70
>

```

MMICT&BM	MM INSTITUTE OF COMPUTER TECHNOLOGY & BUSINESS MANAGEMENT-MULLANA	LABORATORY MANUAL
	PRACTICAL EXPERIMENT INSTRUCTION SHEET	
	Title: Scoping Rules in R	
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LABORATORY: R Programming	Semester-VI	Page: 20 of 47

Aim: Scoping Rules in R

- Demonstrate local and global variables using functions.

Experiment: Write an R program to demonstrate the difference between local and global variables by defining a variable outside a function and modifying it inside the function. Display the values of the variable before and after function execution.

Solution:

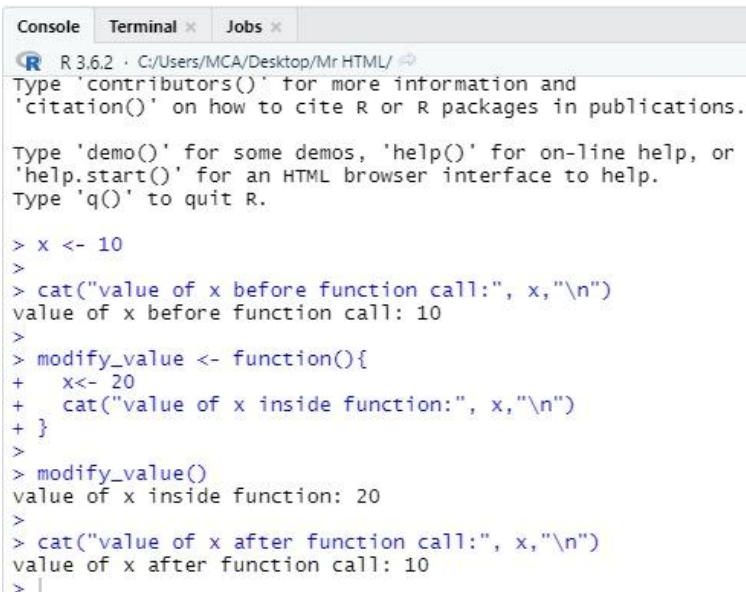
```
x <- 10

cat("value of x before function call:", x, "\n")

modify_value <- function(){
  x<- 20
  cat("value of x inside function:", x, "\n")
}

modify_value()

cat("value of x after function call:", x, "\n")
```

Output:


```
R 3.6.2 · C:/Users/MCA/Desktop/Mr HTML/
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

> x <- 10
>
> cat("value of x before function call:", x, "\n")
value of x before function call: 10
>
> modify_value <- function(){
+   x<- 20
+   cat("value of x inside function:", x, "\n")
+ }
>
> modify_value()
value of x inside function: 20
>
> cat("value of x after function call:", x, "\n")
value of x after function call: 10
> |
```

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Aim: Apply Family Functions

- Use apply(), lapply(), sapply(), and tapply().

Experiment: Use apply(), lapply(), and sapply() on a matrix and compare the results.

Solution:

```

m <- matrix(1:9, nrow = 3)

cat("Matrix:\n")

print(m)

cat("\nUsing apply() - column sums:\n")

print(apply(m, 2, sum))

cat("\nUsing lapply() - column sums:\n")

print(lapply(as.data.frame(m), sum))

cat("\nUsing sapply() - column sums:\n")

print(sapply(as.data.frame(m), sum))

```

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Output:

```

Console Terminal × Background Jobs ×
R - R4.5.2 - ~/
> m <- matrix(1:9, nrow = 3)
>
> cat("Matrix:\n")
Matrix:
> print(m)
      [,1] [,2] [,3]
[1,]    1    4    7
[2,]    2    5    8
[3,]    3    6    9
>
> cat("\nUsing apply() - column sums:\n")

Using apply() - column sums:
> print(apply(m, 2, sum))
[1]  6 15 24
>
> cat("\nUsing lapply() - column sums:\n")

Using lapply() - column sums:
> print(lapply(as.data.frame(m), sum))
$V1
[1] 6

$V2
[1] 15

$V3
[1] 24
>
> cat("\nUsing sapply() - column sums:\n")

Using sapply() - column sums:
> print(sapply(as.data.frame(m), sum))
V1 V2 V3
 6 15 24
> |

```

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	Title: Write a program to display the current system date and time and pause execution for 5 seconds.	
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Aim: Working with Date, Time, and Sleep Functions

Handle system time, dates, and implement delays using Sys.sleep().

Experiment: Write a program to display the current system date and time and pause execution for 5 seconds.

Solution:

```
# Display the current system date and time in readable format
```

```
cat(
```

```
  "Current system date and time:",
```

```
  format(Sys.time(), "%Y-%m-%d %H:%M:%S"),
```

```
  "\n"
```

```
)
```

```
# Pause execution for 5 seconds
```

```
Sys.sleep(5)
```

```
# Message after pause
```

```
cat("Program resumed after 5 seconds.\n")
```

Output:

```

14:1 (Top Level) R Script
Console Terminal Jobs
R 3.6.2 · C:/Users/MCA/Desktop/Mr HTML/
> # Display the current system date and time in readable format
> cat(
+   "Current system date and time:",
+   format(Sys.time(), "%Y-%m-%d %H:%M:%S"),
+   "\n"
+ )
Current system date and time: 2026-02-02 09:51:18
> # Pause execution for 5 seconds
> Sys.sleep(5)
|

```


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	Title: Read a CSV file into R and display summary statistics of the data.	
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Aim: Reading and Writing Data Files

- Read/write CSV and text files using read.csv() and write.csv().

Experiment: Read a CSV file into R and display summary statistics of the data.

Solution:

```
data <- read.csv("C:/Users/MCA/Desktop/r prog/data.csv",
                header = TRUE,
                stringsAsFactors = FALSE)
```

```
head(data)
```

```
str(data)
```

```
summary(data)
```

```
numeric_data <- data[sapply(data, is.numeric)]
```

```
apply(numeric_data, 2, function(x) {
```

```
  c(
```

```
    Min  = min(x, na.rm = TRUE),
```

```
    Max  = max(x, na.rm = TRUE),
```

```
    Mean = mean(x, na.rm = TRUE),
```

```
    Median = median(x, na.rm = TRUE),
```

```
    SD   = sd(x, na.rm = TRUE)
```

```
  )
```

```
})
```

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Output:

```

Console Terminal x Jobs x
R 3.6.2 · C:/Users/MCA/Desktop/Mr HTML/

>
> apply(numeric_data, 2, function(x) {
+   c(
+     Min    = min(x, na.rm = TRUE),
+     Max    = max(x, na.rm = TRUE),
+     Mean   = mean(x, na.rm = TRUE),
+     Median = median(x, na.rm = TRUE),
+     SD     = sd(x, na.rm = TRUE)
+   )
+ })
      x345    x4356
Min    3567.00  647.000
Max    54789.00 6474.000
Mean   37678.00 4198.333
Median 54678.00 5474.000
SD     29541.04 3115.923
> data <- read.csv("C:/Users/MCA/Desktop/r prog/data.c
+               header = TRUE,
+               stringsAsFactors = FALSE)
>
> head(data)
  x345 x4356
1  3567   647
2 54789  6474
3 54678  5474
>
> str(data)
'data.frame':   3 obs. of  2 variables:
 $ x345 : int   3567 54789 54678
 $ x4356: int    647 6474 5474
>
> summary(data)
      x345      x4356
Min.   : 3567   Min.   : 647
1st Qu.:29123   1st Qu.:3060
Median :54678   Median :5474
Mean   :37678   Mean   :4198
3rd Qu.:54734   3rd Qu.:5974
Max.   :54789   Max.   :6474
>
> numeric_data <- data[sapply(data, is.numeric)]
>
> apply(numeric_data, 2, function(x) {
+   c(
+     Min    = min(x, na.rm = TRUE),
+     Max    = max(x, na.rm = TRUE),
+     Mean   = mean(x, na.rm = TRUE),
+     Median = median(x, na.rm = TRUE),
+     SD     = sd(x, na.rm = TRUE)
+   )
+ })
      x345    x4356
Min    3567.00  647.000
Max    54789.00 6474.000
Mean   37678.00 4198.333
Median 54678.00 5474.000
SD     29541.04 3115.923
> |

```

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	Title: Create an R object, save it to a file, and reload it.	
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Aim: Saving and Loading R Data Objects

Use save(), load(), and RData files.

Experiment: Create an R object, save it to a file, and reload it.

Solution:

```
# Create an R object

numbers <- sample(1:50, 100, replace = TRUE)

# Save the object to an RData file

save(numbers, file = "numbers.RData")

# Remove the object from the environment

rm(numbers)

# Load the object back into R

load("numbers.RData")

# Display the reloaded object

print(numbers)
```

Output:

```
1:1 (Top Level) R Script
Console Terminal Background Jobs
R 4.5.2 ~ /
> 
> # Create an R object
> numbers <- sample(1:50, 100, replace = TRUE)
> 
> # Save the object to an RData file
> save(numbers, file = "numbers.RData")
> 
> # Remove the object from the environment
> rm(numbers)
> 
> # Load the object back into R
> load("numbers.RData")
> 
> # Display the reloaded object
> print(numbers)
[1] 33 2 9 32 41 4 49 26 46 47 16 25 6 45 26 7 46 18 40 7 50 33 4 38 22 6 7 4 40 2 8 15 6 30 34 38
[37] 8 31 48 8 45 38 15 33 24 49 28 46 20 24 17 28 35 6 21 29 8 42 23 38 46 32 7 38 43 29 2 18 27 49 21 38
[73] 17 28 45 49 14 33 48 15 15 7 14 15 12 34 9 42 20 40 21 49 45 35 15 20 33 1 35 38
>
```

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	Title: Generate 100 random numbers between 1 and 50 and store them in a vector.	
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Aim: Generating Data in R

- Generate sequences and random data using built-in functions.

Experiment: Generate 100 random numbers between 1 and 50 and store them in a vector.

Solution:

```
# Generate 100 random numbers between 1 and 50

random_numbers <- sample(1:50, 100, replace = TRUE)

# Display the generated data

print(random_numbers)
```

Output:

The screenshot shows the R Studio interface with the console pane active. The code entered is as follows:

```
> # Generate 100 random numbers between 1 and 50
> random_numbers <- sample(1:50, 100, replace = TRUE)
>
> # Display the generated data
> print(random_numbers)
```

The output displays the first 73 elements of the generated random numbers vector:

```
[1] 30 32 25 48 50 12 45 37 43 46  6 14 30 40 17 48 35 12  9 30 21 25 31  5  1 50 16 26 20 16  1 35 34 19 43 38
[37] 35 13 21 15  1 12 44 23 31 10 41 24 25 47 28 26 15 27 29 25  1  7 12 10 44 47 39 46 39  2 34 15 16 30 23 25
[73] 27 37  8 37 34 40 36  5 12 43 37 33 17 27 28 21 18 39 34 47  6 11 32 42 15 14 38 49
```

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	Title: Generate a regular sequence from 1 to 100 with a step size of 5.	
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Aim: Regular and Random Sequences

Use seq(), rep(), runif(), rnorm().

Experiment: Generate a regular sequence from 1 to 100 with a step size of 5.

Solution :

```
# Generate a regular sequence from 1 to 100 with step size 5
sequence <- seq(from = 1, to = 100, by = 5)

# Display the sequence
print(sequence)
```

Output:

```
7:2 (Top Level)
Console Terminal x Background Jobs x
R 4.5.2 ~ /
>
> # Generate a regular sequence from 1 to 100 with step size 5
> sequence <- seq(from = 1, to = 100, by = 5)
>
> # Display the sequence
> print(sequence)
[1] 1 6 11 16 21 26 31 36 41 46 51 56 61 66 71 76 81 86 91 96
> |
```

MMICT&BM	MM INSTITUTE OF COMPUTER TECHNOLOGY & BUSINESS MANAGEMENT-MULLANA	LABORATORY MANUAL
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	Title: Write a program to identify and replace missing values with the mean.	
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Aim: Handling Missing Data

- Use `is.na()`, `na.omit()`, and `complete.cases()`.

Experiment: Write a program to identify and replace missing values with the mean.

Solution :

```
# data <- read.csv("C:/Users/MCA/Desktop/r prog/data.csv",
  header = TRUE,
  stringsAsFactors = FALSE)

colSums(is.na(data))

for (i in names(data)) {
  if (is.numeric(data[[i]])) {
    data[[i]][is.na(data[[i]])] <- mean(data[[i]], na.rm = TRUE)
  }
}

colSums(is.na(data))

summary(data)
```

Output:


```
R 3.6.2 · C:/Users/MCA/Desktop/Mr HTML/
> colSums(is.na(data))
x345 x4356
0      0
> for (i in names(data)) {
+   if (is.numeric(data[[i]])) {
+     data[[i]][is.na(data[[i]])] <- mean(data[[i]], na.rm = TRUE)
+   }
+ }
> colSums(is.na(data))
x345 x4356
0      0
> |
```

MMICT&BM	MM INSTITUTE OF COMPUTER TECHNOLOGY & BUSINESS MANAGEMENT-MULLANA	LABORATORY MANUAL
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	Title: Write a R program to create a data frame and get the structure, statistical summary and nature of the data of a given data frame.	
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Aim: Creating and Manipulating Objects

Work with vectors, matrices, lists, and data frames.

Experiment: Write a R program to create a data frame and get the structure, statistical summary and nature of the data of a given data frame.

Solution :

```
data <- data.frame(
  ID   = c(1, 2, 3, 4, 5),
  Name = c("Amit", "Riya", "Karan", "Neha", "Vijay"),
  Age  = c(21, 22, 20, 23, 21),
  Marks = c(78, 85, 69, 90, 74),
  Grade = factor(c("B", "A", "C", "A", "B"))
)
print(data)
cat("\nStructure of the Data Frame:\n")
str(data)
cat("\nStatistical Summary:\n")
summary(data)
cat("\nNature of the Data:\n")
sapply(data, class)
```

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Output:

```

Source
Console Terminal Jobs
R 3.6.2 · C:/Users/MCA/Desktop/Mr HTML/
> colSums(is.na(data))
x345 x4356
0 0
> for (i in names(data)) {
+   if (is.numeric(data[[i]])) {
+     data[[i]][is.na(data[[i]])] <- mean(data[[i]], na.rm = TRUE)
+   }
+ }
> colSums(is.na(data))
x345 x4356
0 0
> data <- data.frame(
+   ID = c(1, 2, 3, 4, 5),
+   Name = c("Amit", "Riya", "Karan", "Neha", "Vijay"),
+   Age = c(21, 22, 20, 23, 21),
+   Marks = c(78, 85, 69, 90, 74),
+   Grade = factor(c("B", "A", "C", "A", "B"))
+ )
> print(data)
  ID Name Age Marks Grade
1  1 Amit  21   78     B
2  2 Riya  22   85     A
3  3 Karan 20   69     C
4  4 Neha  23   90     A
5  5 Vijay 21   74     B
> cat("\nStructure of the Data Frame:\n")
Structure of the Data Frame:
> str(data)
'data.frame':   5 obs. of  5 variables:
 $ ID   : num  1 2 3 4 5
 $ Name : Factor w/ 5 levels "Amit","Karan",...: 1 4 2 3 5
 $ Age  : num  21 22 20 23 21
 $ Marks: num  78 85 69 90 74
 $ Grade: Factor w/ 3 levels "A","B","C": 2 1 3 1 2
> cat("\nStatistical summary:\n")
Statistical Summary:
> summary(data)
      ID      Name      Age      Marks      Grade
Min.   :1   Amit :1   Min.   :20.0   Min.   :69.0   A:2
1st Qu.:2   Karan:1   1st Qu.:21.0   1st Qu.:74.0   B:2
Median :3   Neha :1   Median :21.0   Median :78.0   C:1
Mean   :3   Riya :1   Mean   :21.4   Mean   :79.2
3rd Qu.:4   Vijay:1   3rd Qu.:22.0   3rd Qu.:85.0
Max.   :5                      Max.   :23.0   Max.   :90.0
> cat("\nNature of the Data:\n")
Nature of the Data:
> sapply(data, class)
      ID      Name      Age      Marks      Grade
"numeric" "factor" "numeric" "numeric" "factor"
>

```


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	Title: i. Extract the sub matrix whose rows have column value from a given matrix.	
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LABORATORY: R Programming	Semester-VI	Page: 32 of 47

Aim: Indexing and Accessing Object Values

Demonstrate indexing systems for different R objects.

Experiment: Write a R program to create a matrix and perform following operations:

- i. Extract the sub matrix whose rows have column value from a given matrix.
- ii. Convert a matrix to a 1 dimensional array.

Solution 1 :

```
# R Program: Create a matrix and extract submatrix based on column value
```

```
# Step 1: Create a matrix
```

```
mat <- matrix(
```

```
  c(10, 20, 30,
```

```
    40, 50, 60,
```

```
    70, 80, 90,
```

```
    100, 110, 120),
```

```
nrow = 4,
```

```
byrow = TRUE
```

```
)
```

```
cat("Original Matrix:\n")
```

```
print(mat)
```

```
# Step 2: Extract submatrix where column 2 value > 50
```

```
sub_mat <- mat[mat[, 2] > 50, ]
```

```
cat("\nSubmatrix (Rows with Column 2 > 50):\n")
```

```
print(sub_mat)
```

```
)
```

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Output:

```

1:2 (Top Level)
Console Terminal x Background Jobs x
R R 4.5.2 ~ /
> # R Program: Create a matrix and extract submatrix based on column value
>
> # Step 1: Create a matrix
> mat <- matrix(
+   c(10, 20, 30,
+     40, 50, 60,
+     70, 80, 90,
+     100, 110, 120),
+   nrow = 4,
+   byrow = TRUE
+ )
>
> cat("Original Matrix:\n")
Original Matrix:
> print(mat)
      [,1] [,2] [,3]
[1,]   10   20   30
[2,]   40   50   60
[3,]   70   80   90
[4,]  100  110  120
>
> # Step 2: Extract submatrix where column 2 value > 50
> sub_mat <- mat[mat[, 2] > 50, ]
>
> cat("\nSubmatrix (Rows with Column 2 > 50):\n")

Submatrix (Rows with Column 2 > 50):
> print(sub_mat)
      [,1] [,2] [,3]
[1,]   70   80   90
[2,]  100  110  120
> |

```

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Solution 2 :

```
# R Program: Convert a matrix to a 1-dimensional array
```

```
# Step 1: Create a matrix
```

```
mat <- matrix(
```

```
  c(10, 20, 30,
```

```
    40, 50, 60,
```

```
    70, 80, 90,
```

```
    100, 110, 120),
```

```
nrow = 4,
```

```
byrow = TRUE
```

```
)
```

```
cat("Original Matrix:\n")
```

```
print(mat)
```

```
# Step 2: Convert matrix to 1-dimensional array
```

```
one_d_array <- as.vector(mat)
```

```
cat("\n1-Dimensional Array:\n")
```

```
print(one_d_array)
```

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Output:

```

21:1 (Top Level)
Console Terminal x Background Jobs x
R 4.5.2 ~ /
>
> # R Program: Convert a matrix to a 1-dimensional array
>
> # Step 1: Create a matrix
> mat <- matrix(
+   c(10, 20, 30,
+     40, 50, 60,
+     70, 80, 90,
+     100, 110, 120),
+   nrow = 4,
+   byrow = TRUE
+ )
>
> cat("Original Matrix:\n")
Original Matrix:
> print(mat)
      [,1] [,2] [,3]
[1,]   10   20   30
[2,]   40   50   60
[3,]   70   80   90
[4,]  100  110  120
>
> # Step 2: Convert matrix to 1-dimensional array
> one_d_array <- as.vector(mat)
>
> cat("\n1-Dimensional Array:\n")

1-Dimensional Array:
> print(one_d_array)
[1]  10  40  70 100  20  50  80 110  30  60  90 120
> |

```

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Aim: Introduction to S3 and S4 Classes

Create and use simple S3 and S4 objects.

Experiment: Demonstrate indexing on vectors, matrices, and data frames.

Solution :

```
# Create an S3 object
person <- list(name = "Alice", age = 25)

# Assign class
class(person) <- "Person"

# Define a print method for the class
print.Person <- function(obj) {
  cat("Name:", obj$name, "\n")
  cat("Age :", obj$age, "\n")
}

# Use the object
print(person)
```

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Output:

```

R ▾ R 4.5.2 · ~/
>
> # Create an S3 object
> person <- list(name = "Alice", age = 25)
>
> # Assign class
> class(person) <- "Person"
>
> # Define a print method for the class
> print.Person <- function(obj) {
+   cat("Name:", obj$name, "\n")
+   cat("Age :", obj$age, "\n")
+ }
>
> # Use the object
> print(person)
Name: Alice
Age : 25
> |

```

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Aim: Reference Classes and User-Defined Functions**Implement reference classes and arithmetic functions.****Experiment:** Create a reference class to perform basic arithmetic operations.**Solution :**

```

MyClass <- setRefClass("MyClass",
  fields = list(x = "numeric", y = "numeric"),
  methods = list(
    showValues = function() {
      cat("x =", x, ", y =", y, "\n")
    }
  ))

obj <- MyClass$new(x = 10, y = 20)

# Access fields

obj$x

obj$y

obj$showValues()

```

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Output:

```

Console Terminal x Background Jobs x
R 4.5.2 · ~/
> MyClass <- setRefClass("MyClass",
+                       fields = list(x = "numeric", y = "numeric"),
+                       methods = list(
+                           showValues = function() {
+                               cat("x =", x, ", y =", y, "\n")
+                           }
+                       ))
> obj <- MyClass$new(x = 10, y = 20)
> # Access fields
> obj$x
[1] 10
> obj$y
[1] 20
> obj$showValues()
x = 10 , y = 20
> |

```


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Aim: Creating Basic 2D and 3D Plots

- Plot line graphs, scatter plots, and bar charts.

Experiment: Create a scatter plot for two numeric variables.

Solution :

```
# Create numeric variables
```

```
x <- c(5, 10, 15, 20, 25, 30)
```

```
y <- c(7, 12, 18, 22, 28, 35)
```

```
# Create scatter plot
```

```
plot(x, y,
```

```
  main = "Scatter Plot of X vs Y",
```

```
  xlab = "X Values",
```

```
  ylab = "Y Values",
```

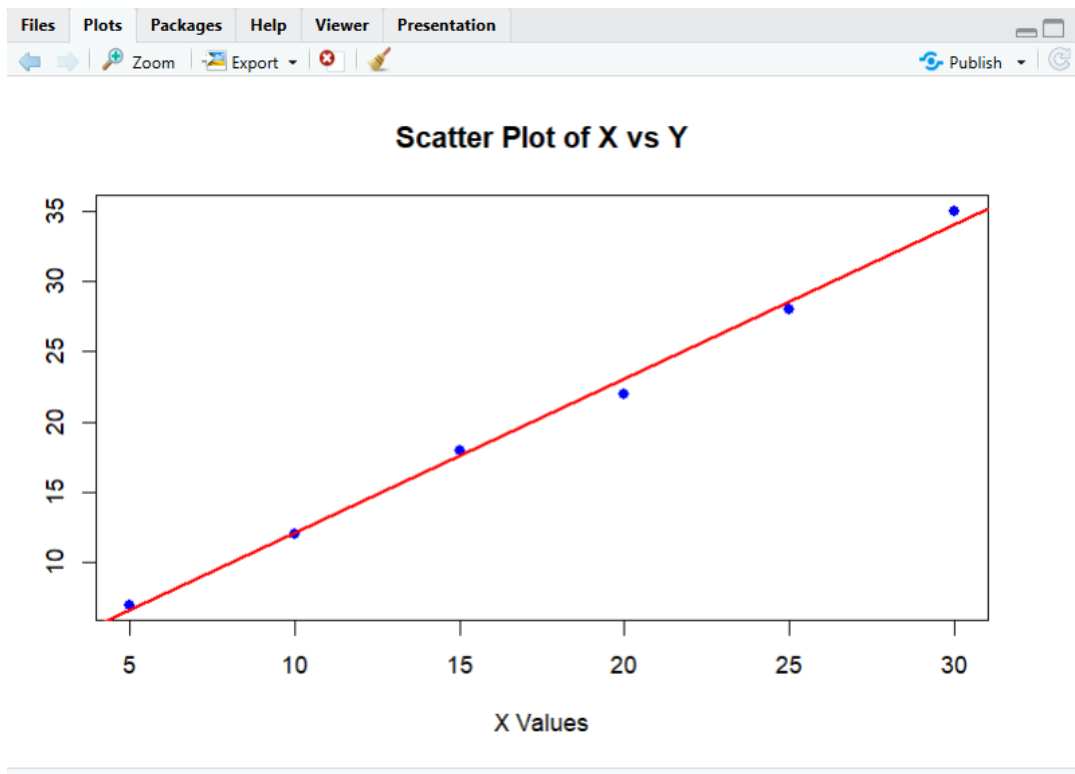
```
  col = "blue",
```

```
  pch = 19)
```

```
# Add regression line
```

```
abline(lm(y ~ x), col = "red", lwd = 2)
```

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Output:

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Aim: Customizing Graphs in R

Add titles, labels, legends, colors, and themes.

Experiment: Customize the plot created in exp16 by adding title, axis labels, and legend.

Solution :

```

x <- c(5, 10, 15, 20, 25, 30)

y <- c(7, 12, 18, 22, 28, 35)

# Custom scatter plot

plot(x, y,

      main = "Customized Scatter Plot of X vs Y",

      xlab = "X Values (Independent Variable)",

      ylab = "Y Values (Dependent Variable)",

      col = "darkblue",

      pch = 19,

      cex = 1.5)

# Add regression line

abline(lm(y ~ x), col = "red", lwd = 2)

# Add legend

legend("topleft",

      legend = c("Data Points", "Trend Line"),

      col = c("darkblue", "red"),

      pch = c(19, NA),

```

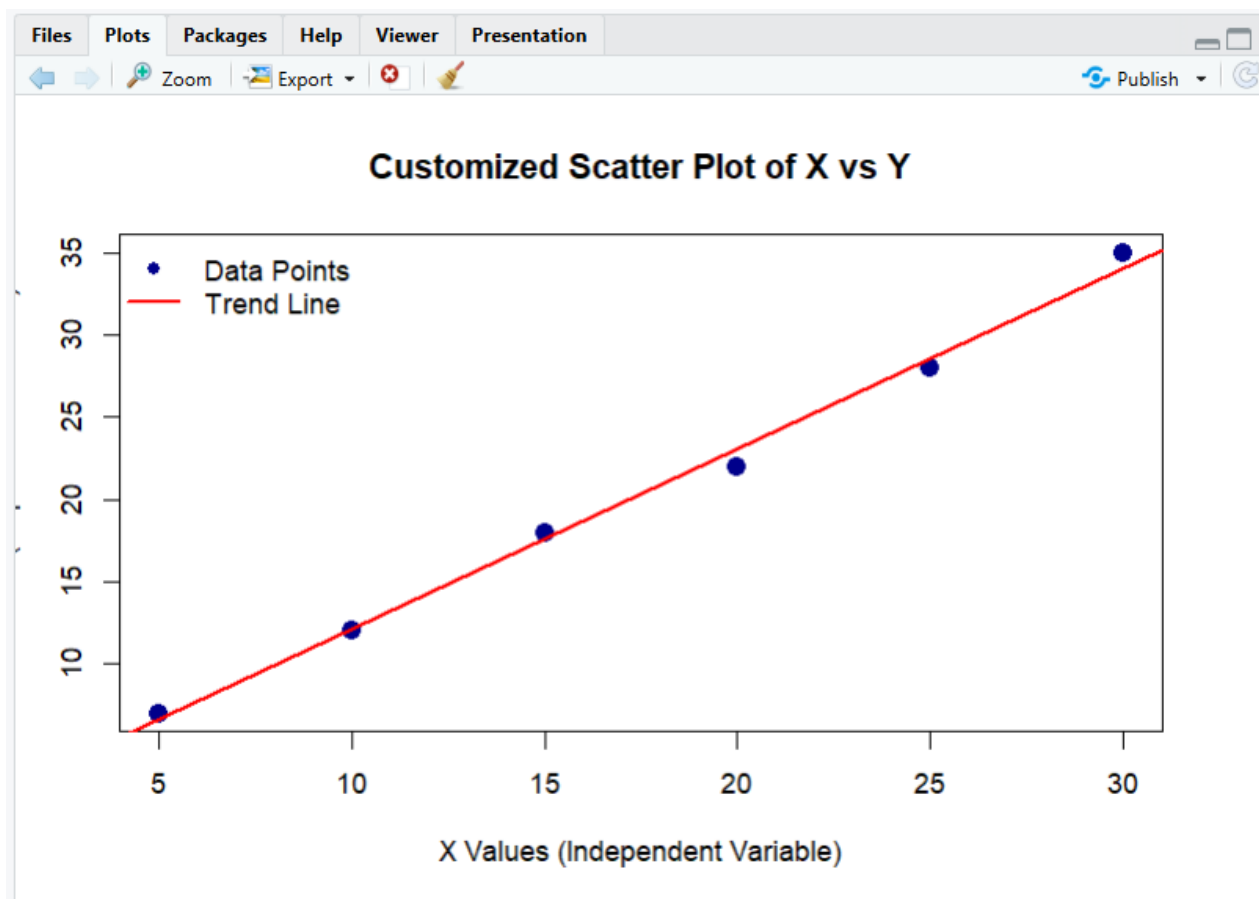
MMICT&BM	MM INSTITUTE OF COMPUTER TECHNOLOGY & BUSINESS MANAGEMENT-MULLANA	LABORATORY MANUAL
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	Title: Customize the plot created in exp16 by adding title, axis labels, and legend	
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lty = c(NA, 1),

lwd = c(NA, 2),

bty = "n"

Output:



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Aim: One-Way ANOVA Using R

Perform and interpret one-way ANOVA.

Experiment: Perform one-way ANOVA on a dataset and interpret the result.

Solution :

Step 1: Create sample data

Suppose we have test scores of students in 3 different teaching methods

```
scores <- c(85, 90, 78, 92, 88, 76, 95, 89, 84, 91, 87, 80)
```

```
method <- factor(c(rep("Method1", 4),
```

```
  rep("Method2", 4),
```

```
  rep("Method3", 4)))
```

Step 2: Combine into a data frame

```
data <- data.frame(scores, method)
```

Step 3: Perform one-way ANOVA

```
anova_result <- aov(scores ~ method, data = data)
```

Step 4: View ANOVA table

```
summary(anova_result)
```

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Output:

```

R 4.5.2 · ~/
Console Terminal × Background Jobs ×
> # Step 1: Create sample data
> # Suppose we have test scores of students in 3 different teaching methods
> scores <- c(85, 90, 78, 92, 88, 76, 95, 89, 84, 91, 87, 80)
> method <- factor(c(rep("Method1", 4),
+                    rep("Method2", 4),
+                    rep("Method3", 4)))
>
> # Step 2: Combine into a data frame
> data <- data.frame(scores, method)
>
> # Step 3: Perform one-way ANOVA
> anova_result <- aov(scores ~ method, data = data)
>
> # Step 4: View ANOVA table
> summary(anova_result)
            Df Sum Sq Mean Sq F value Pr(>F)
method      2    4.5    2.25   0.054  0.947
Residuals   9   371.8   41.31
> |

```

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Aim: Linear Regression Analysis

Implement linear regression, interpret model output, and visualize results.

Experiment: Fit a linear regression model and plot the regression line.

Solution :

```
height <- c(150, 155, 160, 165, 170, 175, 180, 185)
```

```
weight <- c(50, 53, 57, 60, 64, 68, 72, 75)
```

```
# Step 2: Fit linear regression model
```

```
model <- lm(weight ~ height)
```

```
# Step 3: View model summary
```

```
summary(model)
```

```
# Step 4: Plot scatter plot with regression line
```

```
plot(height, weight,
```

```
  main = "Linear Regression: Weight vs Height",
```

```
  xlab = "Height (cm)",
```

```
  ylab = "Weight (kg)",
```

```
  pch = 19, col = "blue")
```

```
abline(model, col = "red", lwd = 2)
```

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